

Chapter-01 Practice Sheet

SECTION A - STEM-AND-LEAF DIAGRAMS

5 medium-to-hard questions

A1 Coin bags and maximum value

A shopkeeper takes 12 bags of coins to the bank. The bags contain the following numbers of coins:

150, 163, 158, 165, 172, 152, 160, 170, 156, 162, 159 and 175.

(a) Represent this information in a stem-and-leaf diagram. Include a key.

Leaves	Stem	
	15	
	16	
	17	

Key: _____

(b) Each bag contains coins of the same value, and the shopkeeper has at least one bag containing coins with dollar values of \$0.10, \$0.25, \$0.50 and \$1.00 only. What is the greatest possible value of all the coins in the 12 bags?

A2 Ferry passengers

Over a 14-day period, data were collected on the number of passengers travelling on two ferries, A and Z. The results are shown below.

Ferry A (14)	Stem	Ferry Z (14)
8 7 6	2	
7 6 4 0	3	0 5 8
8 6 5 3	4	3 4 5 7 7 7
5 3 3	5	0 2 6 6 9

Key: 3 | 5 | 0 represents 53 passengers on A and 50 passengers on Z

(a) How many more passengers travelled on ferry Z than on ferry A?

(b) The cost of a trip on ferry A is \$12.50 and the cost of a trip on ferry Z is \$x. The takings on ferry Z were \$3.30 less than the takings on ferry A over this period. Find x.

(c) Find the least and greatest possible number of days on which the two ferries could have carried exactly the same number of passengers.

A3 Comparing two batsmen

The runs scored by two batsmen in 15 cricket matches last season were:

Batsman P: 53, 41, 57, 38, 41, 37, 59, 48, 52, 39, 47, 36, 37, 44, 59.

Batsman Q: 56, 48, 31, 64, 21, 52, 45, 36, 57, 68, 77, 20, 42, 51, 71.

(a) Show the data in a diagram that allows easy comparison of the two performances.

Batsman P	Stem	Batsman Q
	2	
	3	
	4	
	5	
	6	
	7	

Key: _____

(b) Giving a reason for your answer, decide which of the batsmen performed:

(i) better

(ii) more consistently.

A4 Bird nests: rows of width 5

The total numbers of eggs laid in the nests of two species of bird were recorded over several breeding seasons.

Wrens: 22, 18, 21, 23, 17, 23, 20, 19, 24, 13.

Dunnocks: 28, 24, 23, 19, 30, 27, 22, 25, 22, 17.

(a) Represent the data in a back-to-back stem-and-leaf diagram with rows of width 5.

Wrens (10)	Stem	Dunnocks (10)
	1	
	1	
	2	
	2	
	3	

Key: _____

(b) Given that all of these eggs hatched and that the survival rate for dunnock chicks is 92%, estimate the number of dunnock chicks that survived.

(c) Find the survival rate for the wren chicks, given that 14 did not survive.

A5 Top examination scores

This back-to-back stem-and-leaf diagram shows the percentage scores of the 25 students who were the top performers in an examination.

Girls (12)	Stem	Boys (13)
4 1	8	2
8 6 6	8	5 9
3 2 1 0	9	0 1 3 3 4 4
8 7 7	9	5 6 6 9

Key: 1 | 8 | 2 represents 81% for a girl and 82% for a boy

The 25 students are arranged in a line in the order of their scores. Describe the student in the middle of the line and find the greatest possible number of boys in the line who are not standing next to a girl.

SECTION B - HISTOGRAMS AND FREQUENCY

5 medium-to-hard questions

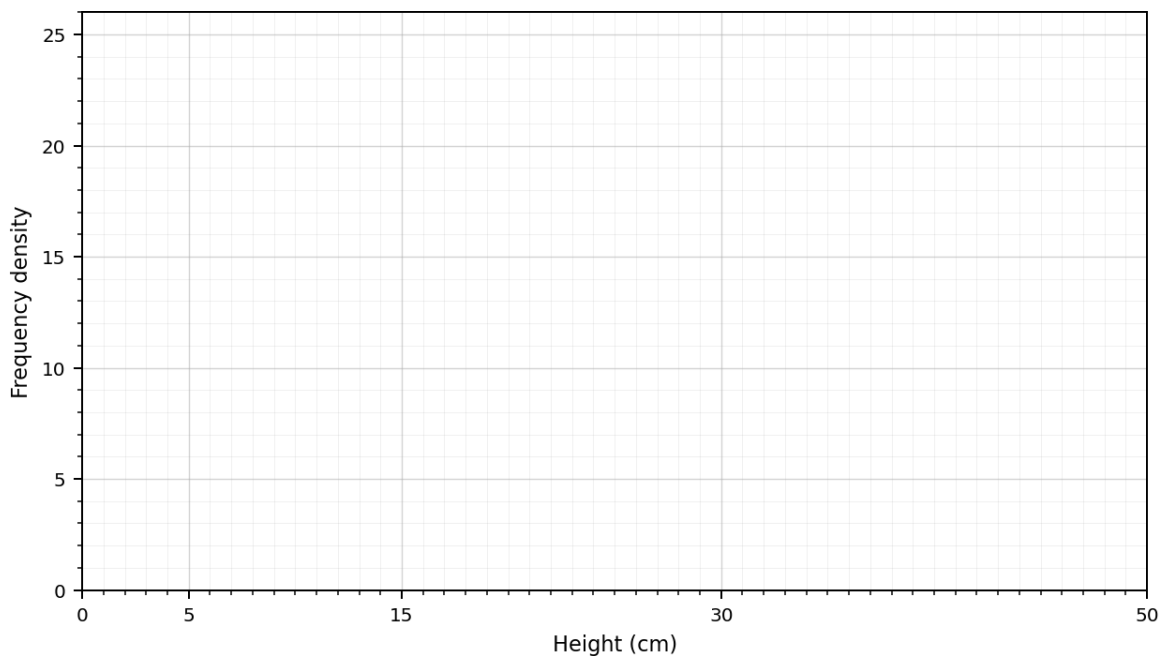
B1 Heights of saplings

The heights of 600 saplings are shown in the following table.

Height (cm)	0-5	5-15	15-30	30-u
No. saplings (f)	64	232	240	64

(a) Suggest a suitable value for u , the upper boundary of the data.

(b) Illustrate the data in a histogram. Use your chosen value of u .



(c) Calculate an estimate of the number of saplings with heights that are:

(i) less than 25 cm

(ii) between 7.5 cm and 19.5 cm.

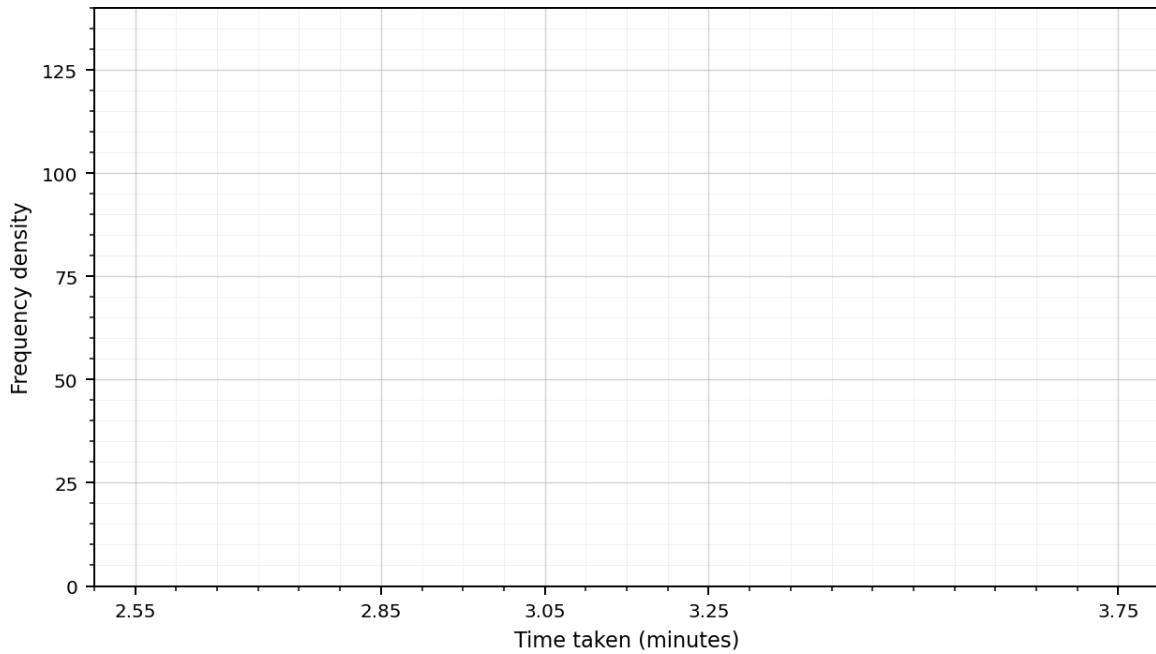
B2 Typing times

Each of the 70 trainees at a secretarial college was asked to type a copy of a particular document. The times taken are shown, correct to the nearest 0.1 minutes, in the table.

Time taken (min)	2.6-2.8	2.9-3.0	3.1-3.2	3.3-3.7
No. trainees (f)	15	25	20	10

(a) Explain why the interval for the first class has a width of 0.3 minutes.

(b) Represent the times taken in a histogram.



(c) Estimate, to the nearest second, the upper boundary of the times taken by the fastest 10 typists.

(d) It is given that 15 trainees took between 3.15 and b minutes. Calculate an estimate for b when:

(i) $b > 3.15$

(ii) $b < 3.15$.

B3 Equal-height histogram columns

The masses, in kilograms, of the animals treated at a veterinary clinic in the past year are illustrated in a histogram. The histogram has four columns of equal height. The table shows the class intervals and the number of animals in two of the classes.

Mass (kg)	3-5	6-12	13-32	33-44
No. animals (f)	a	371	1060	b

- (a)** Find a and b , and show that a total of 2226 animals were treated at the clinic.

- (b)** Calculate an estimate of the lower boundary of the masses of the heaviest 50% of these animals.

B4 Frequency-density and width ratios

The frequency densities of the four classes in a histogram are in the ratio 4 : 3 : 2 : 1. The frequencies of these classes are in the ratio 10 : 15 : 24 : 8.

Find the total width of the histogram, given that the narrowest class interval is represented by a column of width 3 cm.

B5 Missing class limits

The percentage examination scores of 747 students are given in the table.

Score (%)	p-50	51-70	71-80	81-q
No. students (f)	165	240	195	147

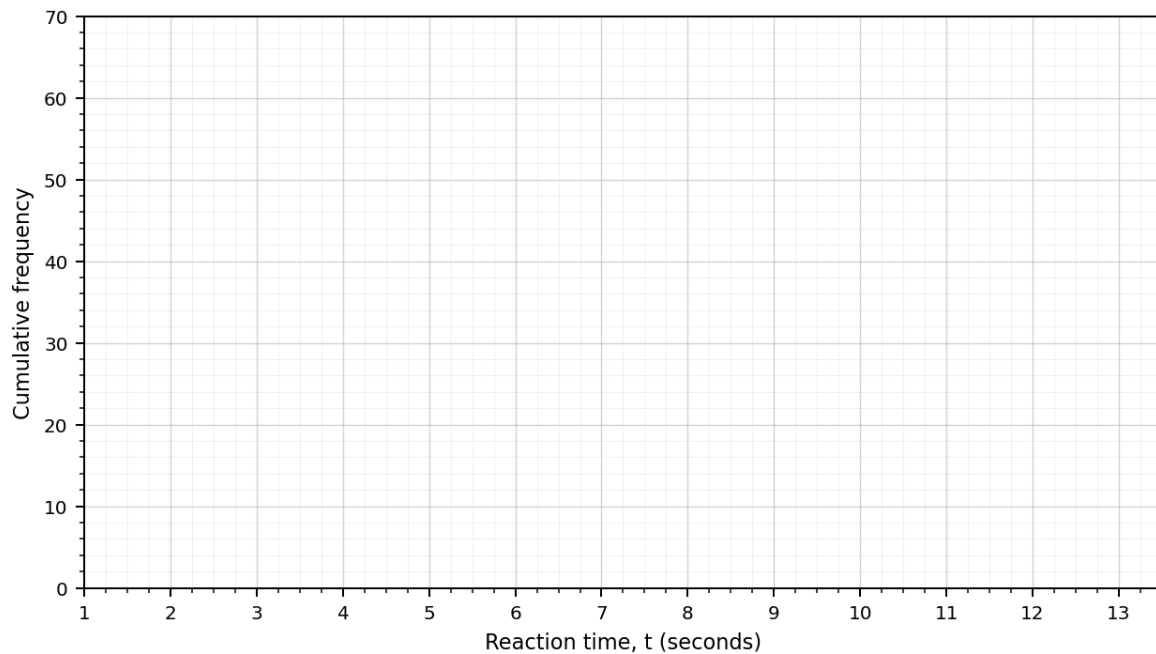
5 medium-to-hard questions

The reaction times, t seconds, of 66 participants were measured in an experiment and presented below.

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Time (t seconds)	No. participants (cf)
$t < 11.0$	62
$t < 13.0$	66

(a) Draw a cumulative frequency polygon to represent the data.



(b) Use your graph to estimate:

(i) the number of participants with reaction times between 5.5 and 7.5 seconds

(ii) the lower boundary of the slowest 20 reaction times.

C2 Widths of books

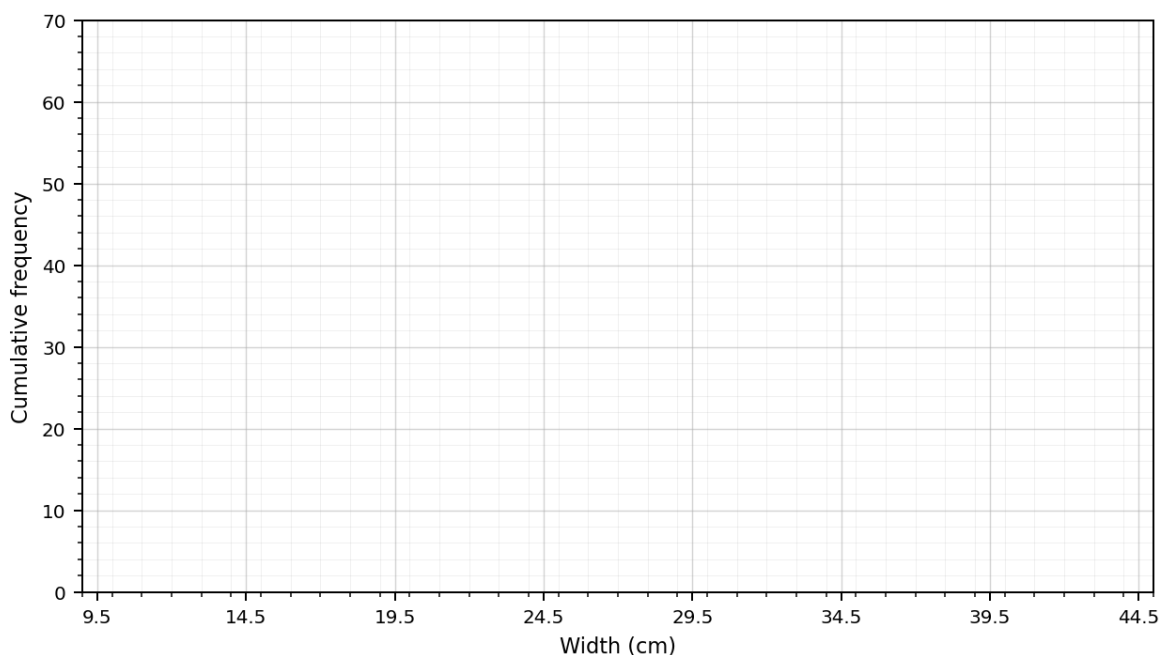
The table shows the widths of the 70 books in one section of a library, given to the nearest centimetre.

Width (cm)	10-14	15-19	20-29	30-39	40-44
No. books (f)	3	13	25	24	5

- (a) Given that the upper boundary of the first class is 14.5 cm, write down the upper boundary of the second class.

- (b) Draw up a cumulative frequency table for the data and construct a cumulative frequency graph.

Upper boundary	9.5	14.5	19.5	29.5	39.5	44.5
Cumulative frequency						



- (c) Use your graph to estimate:

- (i) the number of books that have widths of less than 27 cm

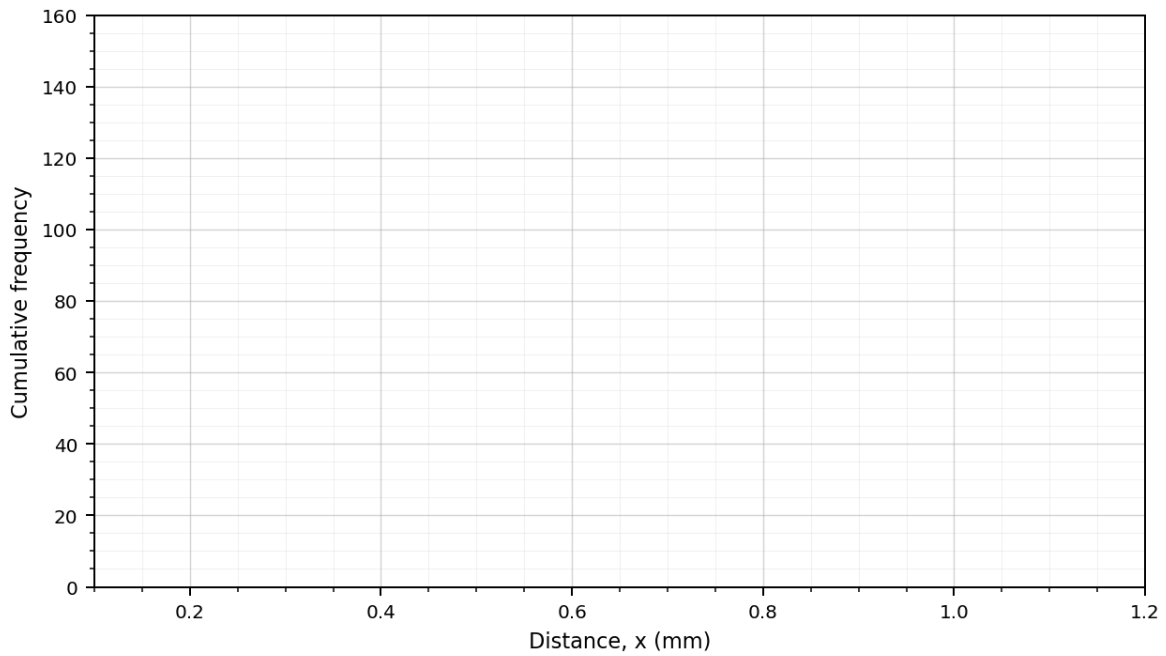
(ii) the widths of the widest 20 books.

C3 Comparing two engine types

Measurements of the distances, x mm, between two moving parts inside car engines were recorded. There were 156 engines of type A and 156 engines of type B.

Distance (x mm)	$x < 0.10$	$x < 0.35$	$x < 0.60$	$x < 0.85$	$x < 1.20$
Engine A (cf)	0	16	84	134	156
Engine B (cf)	0	8	52	120	156

(a) Draw and label two cumulative frequency curves on the same axes.



(b) Use your graphs to estimate:

(i) the number of engines of each type with measurements between 0.30 and 0.70 mm

(ii) the total number of engines with measurements that were less than 0.55 mm.

(c) Both types of engine must be repaired if the distance is more than a certain fixed amount. Given that 16 type A engines need repairing, estimate the number of type B engines that need repairing.

C4 Bank staff journey times

The daily journey times for 80 bank staff to get to work are given in the table.

Time (t min)	t < 10	t < 15	t < 20	t < 25	t < 30	t < 45	t < 60
No. staff (cf)	3	11	24	56	68	76	80

(a) How many staff take between 15 and 45 minutes to get to work?

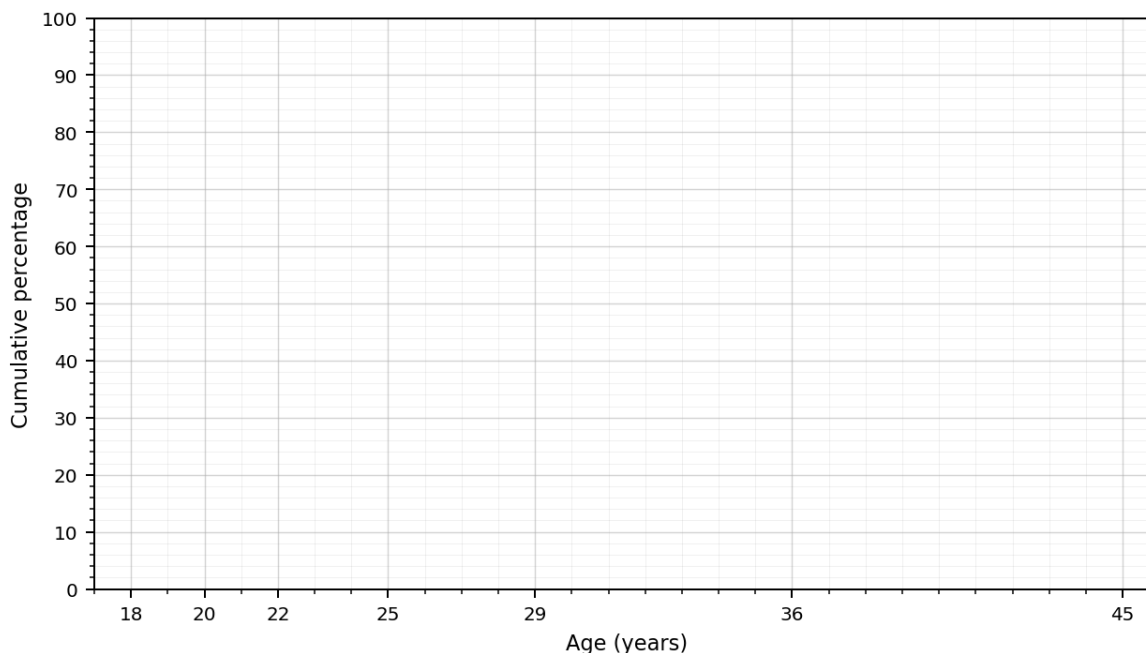
(b) Find the exact number of staff who take $(x + y)/2$ minutes or more to get to work, given that 85% of the staff take less than x minutes and that 70% of the staff take y minutes or more.

C5 University ages and employment

The following table shows the ages of the students currently at a university, given by percentage. Ages are rounded down to the number of whole years.

Age (years)	< 18	18-19	20-21	22-24	25-28	29-35	36-44
Students (%)	0	27	51	11	5	4	2

(a) Represent the data in a percentage cumulative frequency polygon.



(b) The oldest 8% of these students qualify as mature. Use your polygon to estimate the minimum age requirement for a student to be considered mature. Give your answer to the nearest month.

(c) Of the 324 students who are 18-19 years old, 54 are not expected to find employment within 3 months of finishing their course.

(i) Calculate an estimate of the number of current students who are expected to find employment within 3 months of finishing their course.

(ii) What assumptions must be made to justify your calculation in part (c)(i)? Are these assumptions reasonable? Do you expect your estimate to be an overestimate or an underestimate?

SECTION D - END-OF-CHAPTER REVIEW

5 additional medium-to-hard questions

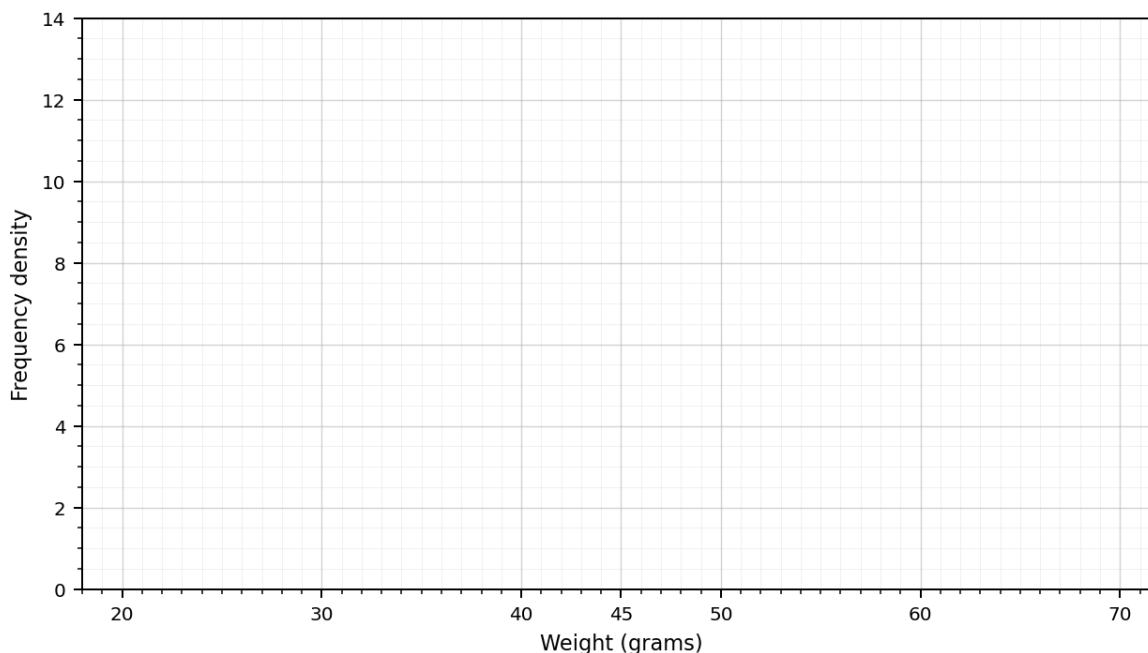
D1 Sausage weights

The weights of 220 sausages are summarised in the table.

Weight (grams)	< 20	< 30	< 40	< 45	< 50	< 60	< 70
Cumulative frequency	0	20	50	100	160	210	220

(i) State how many sausages weighed between 50 g and 60 g. [1]

(ii) On the grid below, draw a histogram to represent the weights of the sausages. [4]



D2 Unknown leaves a and b

The following stem-and-leaf diagram shows eight randomly selected numbers between 2 and 4.

Stem		
	2	1 3 3 a
	3	b 5 6 7

Key: 2 | 1 represents 2.1

Given that $a - b = 7$ and that the sum of the eight numbers correct to the nearest integer is 24, find a and b.

D3 Application download times

Eighty people downloaded a particular application and recorded the time taken for the download to complete.

Download time (min)	< 3	< 5	< 6	< 10
No. downloads (cf)	6	18	66	80

(a) Find the number of downloads that completed in 5 to 6 minutes. [1]

(b) On a histogram, the download times from 5 to 6 minutes are represented by a column of height 9.6 cm.
Find the height of the column that represents the download times of 6 to 10 minutes. [2]

D4 Merging histogram classes

A histogram is drawn with three columns whose widths are in the ratio 1 : 2 : 4. The frequency densities of these classes are in the ratio 16 : 12 : 3, respectively.

(a) Given that the total frequency of the data is 390, find the frequency of each class. [3]

(b) The classes with the two highest frequencies are to be merged and a new histogram drawn. Given that the height of the column representing the merged classes is to be 30 cm, find the correct height for the remaining column. [3]

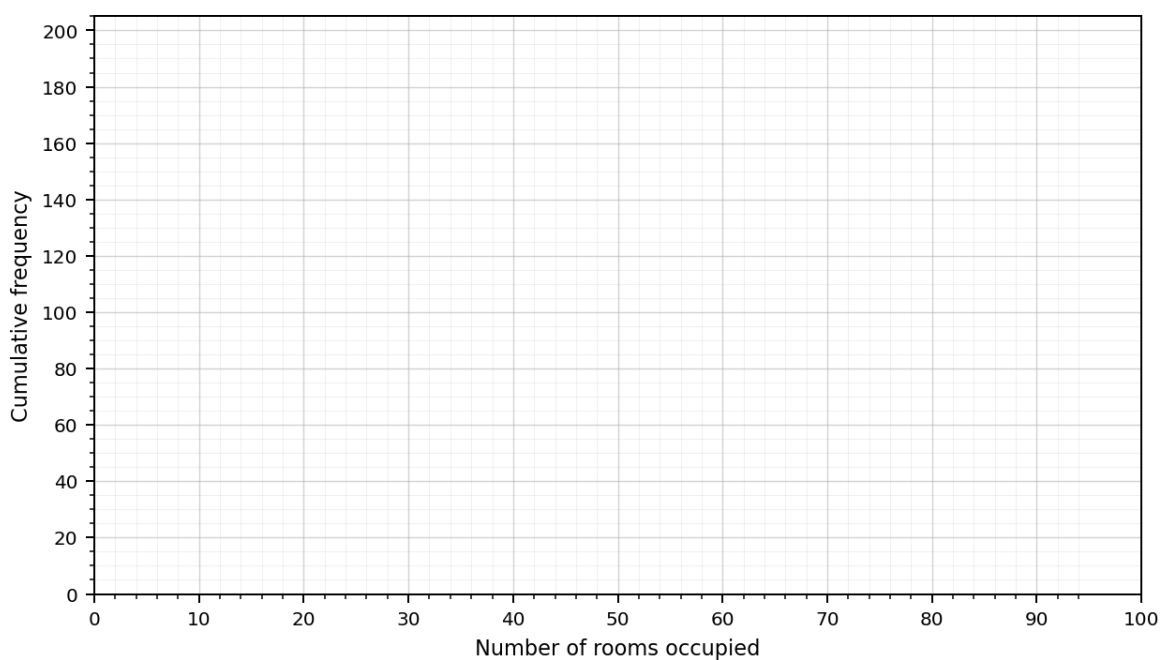
(c) Explain what problems you would encounter if asked to construct a histogram in which the classes with the two lowest frequencies are to be merged. [1]

D5 Hotel room occupancy

A hotel has 90 rooms. The table summarises information about the number of rooms occupied each day for a period of 200 days.

Number of rooms occupied	1-20	21-40	41-50	51-60	61-70	71-90
Frequency	10	32	62	50	28	18

(i) Draw a cumulative frequency graph on the grid below to illustrate this information. [4]



(ii) Estimate the number of days when more than 30 rooms were occupied. [2]

(iii) On 75% of the days at most n rooms were occupied. Estimate n . [2]
